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# COMPUTER FORUM

Volume 1 No. 9

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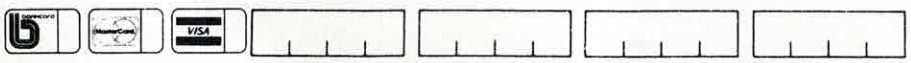
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**MSX and SPECTRAVIDEO**

# COMPUTER FORUM

**Magazine**

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## EDITORIAL

Firstly, let me wish all of you a safe and prosperous 1987 and apologise to those who have been waiting for this issue. We have taken, what we consider, a well-earned break over the holiday period. Traditionally, early January is not a good time to get answers to any questions or to have printing deadlines.

1987 should be a very interesting year for us all. Although MSX is still not being marketed properly, if at all, in this country, we have noticed an upsurge of interest in owners of MSX and SV who have taken their computers out of the cupboard and beginning to realise the potential of these machines.

At present, the future of MSX is based on the strength of the Yen compared to the weakness of our Dollar. The Japanese motor industry is having problems because their products are not selling for the same reason. Let us hope that they have some sway with the people who control their economy. It's pointless talking to our lot!

Happy computing ...

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# MSX2 IS IN AUSTRALIA




The Sony HB-G900P MSX2 computer - the professional.

**We recently had the opportunity of viewing the Sony HB-G900P. This is an MSX2 professional computer used mainly in the video industry.**

The system consists of the main computer and separate keyboard. Across the front of the main 'box' are the power switch, Genlock, Computer, Superimpose and Video selector buttons. Below these is a cartridge slot. Next to the cartridge slot is the built-in disk drive with provision for a second drive next to that. Across the bottom is the reset button, sound volume control, joystick ports and keyboard connector.

The rear panel has another cartridge slot, printer port, tape socket and RGB connector. The other connectors are stereo sound in/out, RS232C, Video interface connector, video reference socket and three switches to select the mode of operation when the computer is used for video control.

Designed for use with the Sony HBI-G900P Videotizer and Laser Disk video system, this unit is only for the professional or very dedicated amateur.

The Videotizer allows the computer to be connected to a range of video equipment for digitizing pictures and superimposing graphics onto a VCR. Although the computer can be used as a stand-alone MSX2, to get the most from this system would also require a Laser Video Disk and possibly a video camera to complete the package.

Retailing at $2634 for the HB-G900P, $1464 for the Videotizer and $336 for the second drive, you may need to check your bankcard limit before racing out to buy one.

## THE ALTERNATIVE

Information to hand indicates that Sony might be looking at bringing in the HB-F700P MSX2 computer. This is the "little brother" of their professional unit and looks very similar.

The main visible difference is the two cartridge slots share the face of the unit with the inbuilt disk drive. A second disk drive can be added externally using the connector on the back. Also across the back are connectors for tape recorder, audio/video, RGB monitor and printer. Our literature showed that the HB-F700P comes with disk software comprising word processor, data base and spreadsheet on disk as well as a mouse which plugs into one of the joystick ports.

No pricing available at present as Sony are still making up their minds whether to market this unit in Australia. Let's hope they do!

## HB-G900P Specifications:

Z80A processor running at 3.58 MHz
Main memory 64K, VRAM 128K.
ROM 48K MSX2 Basic, 16K MSX Disk Basic, 16K Video utility, 14K RS-232C.
CRT controller V9938
Disk drive (built-in): 3.5" 720K formatted, provision for second disk (HBD-G900).
Separate 75 key keyboard
I/O RGB in/out, External sync inputs (BNC), 36 pin Video interface, Audio in/out, Centronics interface, 25 pin RS-232C interface, 2 MSX cartridge slots.

## HB-F700P Main Specifications:

Z80A CPU, 256K RAM, 128K Video RAM, 48K MSX2 BASIC ROM, 16K Disk Basic ROM. 9 Screen modes, up to 512 colours.

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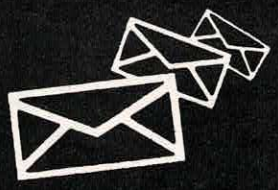
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# MAILBOX



*This page is for your comments and queries. If you have a specific problem or have solved one, we would like to hear about it.*

*If you require an answer before we publish it, please enclose a stamped self-addressed envelope.*

## SV DISK HELP

I have just developed a crushing problem with my Spectravideo system emanating, I think, from the SV902 Disk drive. Periodically, but unpredictably, the logged disk develops a "Bad Allocation Table" error, which as far as I can see "writes off" that disk. It does not happen consistently, or during any particular operations, but is nevertheless devastating - especially now that I have lost my main back-up disk in this manner.

The error message is not mentioned in any lists I have seen published. I have seen only the sketchiest of Spectravideo-published information about the structure and manipulation of the File Allocation Table itself. Is the error fatal? Are my disks now useless?

Christopher Ryan
Nunawading, VIC

*Your disks are not totally useless, just the information they contain - you can re-format them for further use. The Bad Allocation Table error (for all intents and purposes) is FATAL. The information contained on the tracks can be recovered to some extent by using the DSKI$ and DSKO$ commands and a lot of patience and Sherlocking.*

*The error is usually caused by changing disks in the drive without using a FILES command to load the FAT (File Allocation Table) into memory for that disk. For example - loading a file from disk A and then swapping disks requires that a FILES command be done on disk B before the file is saved otherwise the FAT of disk A will be written to disk B. You can get away with it sometimes but not always.*

## SHORT MEMORY

A Disk drive feature in an MSX magazine says that to use MSX-DOS the full 64K RAM must be available and suggests using a 32K RAM expansion cartridge. Although my Sony HB-75AS is listed as having 64K, it signs on at only 28815.

Also, programs such as Monopoly, Scrabble and Cluedo (Virgin) are in either 48K or 64K versions. I assume these would not run on this computer. Nowhere have I seen a 32K RAM cartridge advertised and consider the listed 64K for the HB-75AS to be misleading. What is the position?

Gordon Browell
Biggenden, Qld

*The Sony Hit Bit does, in fact, have 64K RAM fitted. The sign-on message only displays the amount of user RAM available when using MSX Basic in ROM (which occupies the other 32K. Programs written in Basic can only be used with the ROM and therefore must not exceed the amount of user RAM. Machine code programs, on the other hand, do not necessarily need ROM and can be larger than 32K. This is usually marked on the packet. The only MSX machines which need a 32K RAM pack to run this software are the PX-7 and CX5M.*

## BSW PROBLEMS

I now have the SVI707 disk drive with CP/M and MSX-DOS and am very pleased with it except for one major problem. I own a Spectravideo 728 and I have been using Bank Street Writer with a cassette. Since the main reason for purchasing the disk was for word processing, I was shocked to find that I could not get the disk to work with the BSW cartridge. If BSW powers up it does not recognise the existence of the disk drive and if the disk is turned on first, disk Basic boots up. My local dealer knows of the problem but not the cure.

John Knight
Cranbourne, VIC

*The Bank Street Writer cartridge sits in the area &H4000 to &H7FFF and clashes with the disk controller ROM. This is not a problem on some MSX computers which look for a program cartridge first. The SV728 however, looks for CP/M first thereby bypassing the cartridge and booting whatever system is in the drive. Have you tried switching on the disk drive first, but without a disk. The SV738 has no problem running BSW with disks, but the Hit Bit doesn't like it either because of its built-in data base firmware. Has anyone out there found a solution to these problems?*

## BLAGGER HELP

We are really enjoying your magazine. It is lovely to get all this information on the MSX system. Do you have any advice on how to get Blagger out of the first building? We can't get him out of the bank no matter how hard we try!

Mrs M Haack
Baulkham Hills, NSW

*Without giving too much away, Blagger must collect all five keys and then go back to the safe. We found it easiest to collect the top key first followed by the bottom two before getting the keys in the chamber with the blue platforms.*

## LIGHT PEN HELP

I have purchased a Sanyo Light Pen unit (MLP-001) to connect to my Sony Hit Bit computer. All the functions work well but I can't get it to do the graphics on the Sony plotter printer. I asked various people in the game and they all said it would do it, but all I get is a printout of
14
0256
0256 etc.
Could there be a fault in the unit itself or does there have to be a program input first? Hope you can help.

Andrew Barry
Carnarvon, WA

*When we reviewed the Sanyo Light Pen unit our printer was 'on strike' and we were unable to test the printer functions. Have you asked the dealer who supplied the unit to test it? Do any Light Pen owners have the same problem?*

## MSX2 QUESTIONS

I read with interest Ross Brisbane's article entitled "MSX1 & MSX2 Questions Answered" (Issue 6 page 4). I was intrigued that his answers were "... based on using an MSX2 Sony Hit Bit".

The only Sony MSX2 computer that I know about is their HB-G900P. I already own a Sony (MSX1) Hit Bit HB-75AS and have been most impressed with it. I am thinking of upgrading to MSX2 - with the Yamaha CX5MII/128 being the favourite at this time. If however, Sony have an MSX2 Hit Bit on the market in addition to their HB- G900P (as Ross Brisbane suggests), then I would very much like to have details of it.

Bryan Connell
Scarborough, WA

*It appears that the name HIT BIT is used by Sony for their complete range of MSX computers, be they MSX1 or MSX2. As far as we are able to ascertain, the new Yamaha is MSX1 but with some features of MSX2 - such as 128k RAM and 80 column capability.*


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# UNDERSTANDING MACHINE CODE

Due to circumstances beyond our control, it seems that we will not be able to publish the intended Machine Code Tutorial. To assist those with an interest in the subject, we commence a series of articles which will form an introduction to this type of programming.

There are many excellent books available on machine code, however, most that we have seen appear to start at about chapter 3 - that is, chapter 1 does not start off with the basics.

## MACHINE CODE AND HARDWARE

Writing programs in Basic does not require very much knowledge of the workings of a computer and the components which make up that computer. The rules by which you write a program are dictated by the Interpreter - a machine code program which converts your Basic program into instructions which the computer can understand.

Machine code programming, on the other hand, requires a working knowledge of the Central Processing Unit (CPU) and a good understanding of binary maths and logic. Logic dictates that something IS or it IS NOT. Logic is what you use in an IF/THEN statement. IF something IS equal THEN ... IF something IS NOT equal THEN ...

The CPU is a single chip which is very much like a computer with a very small amount of memory - only 26 bytes called registers. These are -

**FLAGS REGISTER (F)** - each bit of this byte is significant and is used for the Sign (S), Zero (Z), Auxiliary Carry (AC), Parity/Overflow (P/O), Subtract (N) and Carry (C).
**ACCUMULATOR (A)** - main one byte data register.
**SECONDARY ACCUMULATORS** consisting of -
B and C REGISTERS - two eight bit registers which can be used to store a 16 bit value.
D and E REGISTERS - two eight bit registers which can be used to store a 16 bit value.
H and L REGISTERS - two eight bit registers which can be used to store a 16 bit value.

All of the above registers have 'complimentary registers' - F', A', B', C', D', E', H', L'

STACK POINTER (SP) - 16 bit address register
INDEX REGISTER (IX) - 16 bit register
INDEX REGISTER (IY) - 16 bit register
INTERRUPT VECTOR (I) - 8 bit register
REFRESH REGISTER (R) - 8 bit register

These are used to hold instructions and information fetched from the RAM of the computer it controls.

## COMPARING MACHINE CODE WITH BASIC

The fundamental difference between a machine code program and one written in Basic is that Basic uses line numbers and variables whereas machine code uses memory locations. We are all familiar with the way variables are assigned values in Basic. The command X=10 means that, until changed, X will have the value of 10. In a machine code program, values are stored in registers and memory locations rather than variables.

The nearest machine code instruction equivalent to the Basic command LET is the LOAD REGISTER command. In a Basic program we would store the value 8 in variable X with the command X=8 or LET X=8. In machine code we could store the value 8 in register A using LD A,8 which means LOAD REGISTER A WITH 8.

One of the easiest ways to understand a new subject is to make comparisons with what we already know. This can be error prone, but nothing ventured ...

Let us compare a Basic program with the steps required for a machine code program to perform the same function.

```
10 INPUT A$
20 PRINT A$
```

This program waits for a key to be pressed and, until the ENTER key is pressed, that character is added to the value of A$. When ENTER is pressed, the value of A$ is displayed on the screen. The machine code program would need to

do something like this -

1. Set aside an area of memory for the contents of A$ and remember the starting point.
2. As the keyboard is a matrix with each key positioned at the intersection of a row and column, output a signal to each row and sample the signals at each column to see if a key is pressed.

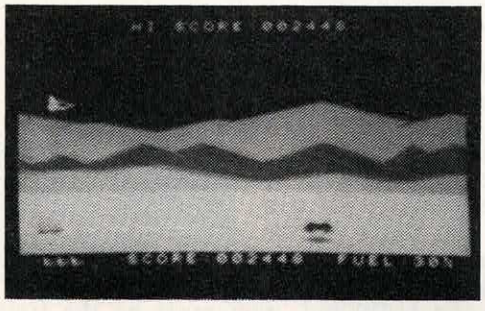
3. Is a key pressed? No - goto 2.
4. Is it ENTER? No - place it in the area for A$ at the current location. Add 1 to the location number and add 1 to the total counter. Go to 2.
5. The ENTER key has been pressed, so we can print the value of A$.
6. Look at the first character of A$.
7. Check the memory locations which contain the current location of the cursor on the screen.
8. Send the first character of A$ to the cursor position.
9. Add 1 to the cursor position.
10. Add 1 to the counter for reading A$.
11. Is the counter equal to the length of A$.
12. No - then look at the next character of A$. Goto 7.
13. All the characters have been printed so go back to the ROM and display the Ok message.

As you can see, the machine code program goes into a lot more detail than the equivalent Basic one. This is because each BASIC command in our program has a machine code routine stored in ROM which performs the steps listed in our machine code version.

The difference is similar to saying "Drive the car" to someone who can drive and "I will show you how to drive the car" to someone who cannot.

# *Software NEWS & REVIEWS*

### U.F.O. (MSX Cassette)
### by FDS Software



Action is fast and furious in this machine code game. You control a space fighter and must destroy the UFOs. Watch out for the alien unit moving across the bottom of the screen - this will fire 'smart bombs'.

Pressing the trigger or space bar fires plasma bolts and missiles are launched by pushing the joystick to the right or pressing the second fire button on an MSX joystick. If you survive the first attack, you are caught in a meteor storm which you must avoid or destroy to attack more UFOs.

Refuel when the fuel truck appears on screen - but watch out, it can back into you if you do not refuel in time. This game, which requires 16K RAM, has smooth scrolling and fast response with good sound effects.

Publisher: Forrest Data Services
Recommended price: $19.95


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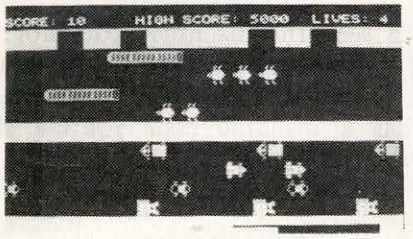
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## GAME PACK 6 (MSX & SV Cassette) by FDS Software

This tape has four titles - HOPPER, SPACE WAR, OTHELLO CHALLENGER and PAINTER




**Hopper** - Hoppy must cross the road and the river to get home. He must avoid the traffic and cross the river without getting wet (strange how frogs can't swim in computer games!). On later levels he must watch out for snakes and crocodiles which will block his path. Has 8 levels. Very enjoyable game.

**Space War** - This is a two player space duel game where each player controls a space craft which can move freely around and shoot at his opponent while trying to avoid being shot himself. First to score 10 wins.

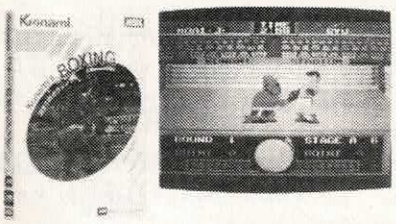
**Othello Challenger** - A 1 or 2 player computer version of the board game where each player must place his tokens to outflank and capture his opponent's pieces. Good strategy game.

**Painter** - A useful utility to explore the graphics capabilities of your computer. This is an on-screen artist which draws lines and shapes, colours them in and provides special effects such as random dots, blocks, circles etc. Has screen save and load option.

Publisher: Forrest Data Services
Recommended price: $15.00

***We review some of the latest MSX cartridges and tapes - courtesy of LPG Software in Hobart.***

## KONAMI BOXING (MSX Cartridge)

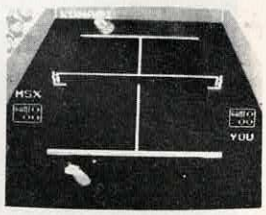



The formidable heavyweight boxer, Fighting-Ryu is in the ring for a series of bouts with the world's leading boxers. He goes the distance of three rounds or wins by a knockout against the best of them. There's More-Betteryo-Ali from America (?), Red-Wolf from England, Sanchess of Italy, Moai-King the Chinese China-Khan and the world giant Moai Jnr. It's knockout or be knocked out in this series of blocks, uppercuts and the famous jabbing, bobbing and weaving.

1 or 2 player, average music, good graphics - great for boxing fans. Hard to learn to control the boxer.

Publisher: Konami
Price: $49.95

## KONAMI PING-PONG (MSX Cartridge)

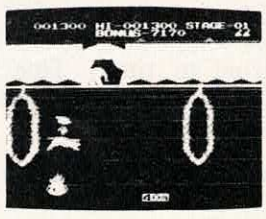



This is the competition of the year. It's the Ping Pong championships and the winner is the person who wins at least two of the three games. To win a game you have to be able to serve, drive, spin and smash. Using forehand and backhand, you must be the first to win 21 points in order to take the set. It's fast and extremely skillful. Test your reflexes against the computer, someone else or make it really hard and play against yourself. Good luck!

1 or 2 player, good graphics and sound - realistic!

Publisher: Konami
Price: $49.95

## CIRCUS CHARLIE (MSX Cartridge)



Circus Charlie is performing in the Big Top! He must jump through wheels of fire, balance on a tight-rope whilst avoiding oncoming monkeys, perform on balancing balls, do trick-riding and lastly, Circus

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# BASIC TUTORIAL - Part 9

## The ON STOP command

This command is used to prevent a running program being interrupted by the user pressing the CTRL/STOP keys. It takes the form -

```
ON STOP GOSUB n
```

where n is the line number of the routine for handling this situation. It is activated by a STOP ON command and de-activated by STOP OFF.

Example 1:

```
10 ON STOP GOSUB 100
20 STOP ON
30 PRINT "COMPUTER FORUM"
40 GOTO 30
100 PRINT "GO AWAY"
110 STOP OFF
120 RETURN
```

This example uses STOP OFF in line 110 to de-activate the ON STOP command. If this had not been included, you would have had to switch off the computer to regain command, as in example 2 -

Example 2:

```
10 ON STOP GOSUB 100
20 STOP ON
30 PRINT "COMPUTER FORUM"
40 GOTO 30
100 PRINT "GO AWAY"
110 PRINT "I AM BUSY PRINTING COMPUTER FORUM"
120 RETURN
```

## The ON KEY command

This command is used to control the operation of a program by use of the function keys F1 to F10. It takes the form -

```
ON KEY GOSUB a,b,c,d,e,f,g,h,i,j
```

where a to j are the line numbers of the routines to jump to for each of the 10 function keys. The line number for each key must be placed in the correct order - the line to jump to for F5 should go where the letter e now is.

Each function key is activated by the command KEY (n) ON where n is the number of the key to activate, and de-activated by the command KEY (n) OFF.

Example 1:

```
10 ON KEY GOSUB 70,80,90
20 KEY (1) ON
30 KEY (2) ON
40 KEY (3) ON
50 PRINT "FUNCTION KEY TEST"
60 GOTO 50
70 PRINT "F1 PRESSED":RETURN
80 PRINT "F2 PRESSED":RETURN
90 PRINT "F3 PRESSED":RETURN
```

If we wanted to check for any function key pressed, we can have each key jump to the same routine -

Example 2:

```
10 ON KEY GOSUB 70,70,70,70,70,70,70,70,70,70
20 FOR I=1 TO 10
30 KEY (I) ON
40 NEXT I
50 PRINT "FUNCTION KEY TEST"
60 GOTO 50
70 PRINT "FUNCTION KEY PRESSED":RETURN
```

To check for only one key pressed, then the following example could be used. To check for function key 3 pressed -

Example 3:

```
10 ON KEY GOSUB ,,50
20 KEY (3) ON
30 PRINT "FUNCTION KEY TEST"
40 GOTO 30
50 PRINT "F3 PRESSED":RETURN
```

## The KEY LIST command

This command displays the current uses of the function keys F1 - F10. If you type KEY LIST and press ENTER, the following will be displayed (unless the functions have been changed by a program).

```
color                - (function of F1)
auto                 - (function of F2)
goto                 - (function of F3)
list                 - (function of F4)
run + enter          - (function of F5)
color 15,4,4 + enter - (function of F6)
cload"               - (function of F7)
cont + enter         - (function of F8)
list. + enter        - (function of F9 list last line entered or listed)
cls + run + enter    - (function of F10)
```

# PROGRAMMING TECHNIQUES

If you were to take a diploma or degree in computer programming, the chances are that, sometime during the Basic unit, you would be given an assignment which required you to convert degrees fahrenheit to degrees centigrade. This is a fairly simple task, because you are given the formula of centigrade=(fahrenheit - 32) * 5 / 9

The simplest form of program would be -

```
10 F=60
20 C=(F-32)*5/9
30 PRINT C
```

You would then be asked to modify the program to allow for user input -

```
10 INPUT "DEGREES FAHRENHEIT";F
20 C=(F-32)*5/9
30 PRINT F;" DEGREES FAHRENHEIT =";C;" DEGREES CENTIGRADE"
```

Great! If you remembered to include your name and the name of your lecturer in the program and included a flow chart, you probably would get 80% for this assignment. Sounds easy doesn't it. The problem when I was doing it was that while you are trying to do these assignments, you must learn to use their computer which did not have on-screen editing but a combination of keys would insert or delete characters. For the first two assignments, if you made a typing error, it was quicker to delete the program and start again. Coupled with the fact that the printer (albeit a high speed line printer) was shared by about 500 students at a time, and the whole process gets bogged down.

Some procedures that were picked up during all of this, may be of use -

Example - Rounding of decimal numbers (4 down, 5 up)

```
10 INPUT "ENTER A DECIMAL NUMBER";D
20 IF D<>INT(D) THEN D=INT(D+.5)
30 PRINT D
```

Line 20 of this routine checks to see if a whole number has been entered. If it is a decimal, it adds .5 to it and then calculates the integer, or whole number value. eg INT(12.7 + .5) = 13 rounding upwards or INT(12.2 + .5) = 12 rounding down.
Other useful routines were sorting and searching.

# SORTING

One of the main functions of any business oriented software is to sort information. Sorting is done using the ASCII codes which were discussed in part 7.

The following program will test ASCII values of two characters which are

input -

```
10 INPUT "FIRST";A$
20 INPUT "SECOND";B$
30 PRINT "ASCII VALUE OF FIRST =";ASC(A$)
40 PRINT "ASCII VALUE OF SECOND =";ASC(B$)
50 IF A$>B$ THEN PRINT "FIRST > SECOND" ELSE PRINT "SECOND > FIRST"
60 PRINT : GOTO 10
```

Example of a simple sort

```
10 CLEAR 500 : DIM N$(9)
20 FOR I=1 TO 9
30 INPUT"ENTER NAME";N$(I)
40 NEXT I
50 REM SORT INTO ASCENDING ORDER
60 FOR I=1 TO 9
70 FOR J=1 TO 9
80 IF N$(I)<N$(J) THEN SWAP N$(I),N$(J)
90 NEXT J,I
100 FOR I=1 TO 90
110 PRINT N$(I)
120 NEXT I
```

The way in which this works can be demonstrated with this program.

```
10 FOR R=1 TO 4:READ N$(R):NEXT R
20 FOR I=1 TO 4:FOR J=1 TO 4
30 IF N$(I)<N$(J) THEN SWAP N$(I),N$(J)
40 PRINT I;" ";J;" ";:FOR L=1 TO 4
50 PRINT N$(L);:NEXT L:PRINT
60 NEXT J
70 NEXT I
80 DATA D,C,B,A
```

The J loop is 'nested' inside the I loop and the value of J changes 4 times for every 1 that I changes. The effect is -

```
Pass 1:  N$(I)=D,N$(J)=D (no swap)
Pass 2:  N$(I)=D,N$(J)=C (no swap)
Pass 3:  N$(I)=D,N$(J)=B (no swap)
Pass 4:  N$(I)=D,N$(J)=A (no swap)
Pass 5:  N$(I)=C,N$(J)=D (C<D so swap) N$(I)=D,N$(J)=C : CDBA
Pass 6:  N$(I)=D,N$(J)=C (no swap)
Pass 7:  N$(I)=D,N$(J)=B (no swap)
Pass 8:  N$(I)=D,N$(J)=A (no swap)
Pass 9:  N$(I)=B,N$(J)=D (B<C so swap) N$(I)=C,N$(J)=B : BDCA
Pass 10: N$(I)=C,N$(J)=D (C<D so swap) N$(I)=D,N$(J)=C : BCDA
Pass 11: N$(I)=D,N$(J)=D (no swap)
Pass 12: N$(I)=D,N$(J)=A (no swap)
Pass 13: N$(I)=A,N$(J)=B (A<B so swap) N$(I)=B,N$(J)=A : ACDB
Pass 14: N$(I)=B,N$(J)=C (B<C so swap) N$(I)=C,N$(J)=B : ABDC
Pass 15: N$(I)=C,N$(J)=D (C<D so swap) N$(I)=D,N$(J)=C : ABCD
Pass 16: N$(I)=D,N$(J)=D (no swap)
```

Sometimes the records you wish to sort will have more than one field per record. If one field is swapped, then all fields in that record must be swapped with it.

Sort on name field, keeping other data with the sorted field -

```
10 CLEAR 1500 : DIM N$(9),S$(9),T$(9)
20 FOR I=1 TO 9
30 INPUT "NAME";N$(I)
40 INPUT "STREET";S$(I)
50 INPUT "TOWN";T$(I)
60 NEXT I
70 REM SORT ON NAME
80 FOR I=1 TO 9
90 FOR J=1 TO 9
100 IF N$(I)<N$(J) THEN SWAPN$(I),N$(J):SWAPS$(I),S$(J):SWAPT$(I),T$(J)
110 NEXT J,I
120 FOR I=1 TO 9
130 PRINT N$(I)
140 PRINT S$(I)
150 PRINT T$(I)
160 PRINT
170 NEXT I
```

The previous example will sort on the name field only. There are times when you will want the same information sorted on a different field.

```
10 CLEAR 1500 : DIM N$(9),S$(9),T$(9)
20 FOR I=1 TO 9
30 INPUT "NAME";N$(I)
40 INPUT "STREET";S$(I)
50 INPUT "TOWN";T$(I)
60 NEXT I
70 REM GET SORT FIELD
80 CLS : PRINT"SORT ON WHICH FIELD"
90 PRINT
100 PRINT "1. NAME"
110 PRINT "2. STREET"
120 PRINT "3. TOWN"
130 INPUT "ENTER 1,2 OR 3";F
140 FOR I=1 TO 9
150 FOR J=1 TO 9
160 ON F GOSUB 250,270,290
170 NEXT J,I
180 FOR I=1 TO 9
190 PRINT N$(I)
200 PRINT S$(I)
210 PRINT T$(I)
220 PRINT
230 NEXT I
240 END
250 IF N$(I)<N$(J) THEN SWAPN$(I),N$(J):SWAPS$(I),S$(J):SWAPT$(I),T$(J)
```

```
260 RETURN
270 IF S$(I)<S$(J) THEN SWAPN$(I),N$(J):SWAPS$(I),S$(J):SWAPT$(I),T$(J)
280 RETURN
290 IF T$(I)<T$(J) THEN SWAPN$(I),N$(J):SWAPS$(I),S$(J):SWAPT$(I),T$(J)
300 RETURN
```

## SEARCHING

There are two ways to search a list of items. The first of these is a SEQUENTIAL search. That is, look at the first item, if it is not the one required, move on to the next and so on.

```
10 CLEAR 500 : DIM N$(9)
20 FOR I=1 TO 9
30 READ N$(I)
40 NEXT I
60 REM SEARCH FOR 'ZAK'
70 FOR I=1 TO 9
80 IF N$(I)="ZAK" THEN PRINT "FOUND ZAK AT POSITION";I
90 NEXT I
100 END
110 DATA DAVE,STEVE,JOHN,ALAN,RON,TOM,MIKE,WALLY,ZAK
```

Because ZAK was the last name on the list, the program had to look at every name on the list before it found ZAK.

To speed up the search, we can use a system called a binary search. The binary search requires that the list is first sorted into ascending order. Then the list is continually divided in half and the name at the mid point is checked for a match.

```
10 CLEAR 500 : DIM N$(9)
20 FOR I=1 TO 9
30 READ N$(I)
40 NEXT I
50 REM SORT INTO ASCENDING ORDER
60 FOR I=1 TO 9
70 FOR J=1 TO 9
80 IF N$(I)<N$(J) THEN SWAP N$(I),N$(J)
90 NEXT J,I
100 LP=1:HP=9
110 MP=INT((HP-LP)/2)+LP
120 IF MP=LP THEN MP=HP
130 IF N$(MP)<"ZAK" THEN LP=MP : GOTO 110
140 IF N$(MP)>"ZAK" THEN HP=MP : GOTO 110
150 PRINT "FOUND ZAK AT POSITION";MP
160 DATA DAVE,STEVE,JOHN,ALAN,RON,TOM,MIKE,WALLY,ZAK
```

How the program works -

| PASS 1 | | PASS 2 | | PASS 3 | | PASS 4 | |
|---|---|---|---|---|---|---|---|
| ALAN | LP | RON | LP | TOM | LP | TOM | LP |
| DAVE | | STEVE | | WALLY | MP | ZAK | MP |
| JOHN | | TOM | MP | ZAK | HP | WALLY | HP |
| MIKE | | WALLY | | | | | |
| RON | MP | ZAK | HP | | | | |
| STEVE | | | | | | | |
| TOM | | | | | | | |
| WALLY | | | | | | | |
| ZAK | HP | | | | | | |

# ERROR MESSAGES

## SYNTAX ERROR

This is probably the most common error message. It simply means that the computer has come across a command which has not been entered into the program correctly. Most of the time this message will be generated by a simple typing error or mispelling. The proper syntax is based on the fact that -

Each line must have a line number.
The first word of each line must be a command.
Commands on the same line must be separated by a colon (:).
Certain commands must be followed by values.
These values must be presented in a certain way - sometimes in brackets, sometimes in quotes (").

## TYPE MISMATCH

This error is caused when the computer is expecting a numeric value and gets a string, or vice versa.

For example -

| INCORRECT | CORRECT |
|---|---|
| PLAY ABC | PLAY "ABC" |
| CHR$("A") | CHR$(X) or CHR$(65) |
| ASC(65) | ASC("A") |
| D="ABC" | D$="ABC" |
| D$=10 | D=10 |

## MISSING OPERAND

This error message is generated by the fact that you have not given the computer the whole story for it to carry out the command. This can occur

when you use an operator (+,-,*,/, etc) without a subsequent operand - eg PRINT 10/

or you have not completed the instruction such as -

POKE 1000,

## ILLEGAL FUNCTION CALL

For MSX and SV computers, this error is usually the result of using graphics commands in a program without previously using the correct SCREEN command.

It can also be caused by using a negative value for a subscript -

PRINT A(-1)

## SUBSCRIPT OUT OF RANGE

This error message is caused by a subscripted variable being used incorrectly, or more likely, a typing error. For example

CHR$(65) being entered as CHR(65)

Subscripted variables are those which are followed by a bracketed number

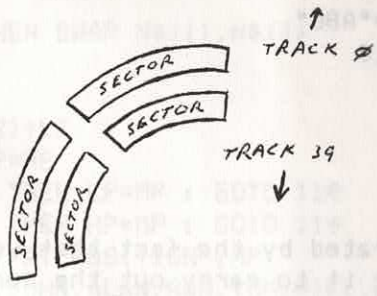
eg A(1),A(2) etc

To use more than ten subscripted variables, an array must be DIMensioned for them. eg DIM A(12)

Even DIMensioning an array can cause this error if the number of the subscript exceeds the array dimension. eg DIM A(12) : A(13)=10

## DISK BASIC

Continuing this month with the basics of Disk Basic, we introduce the way information is stored on the disk. Each disk is divided into tracks (concentric magnetic circles formed on the surface of the disk) which in turn are divided into sectors.



# The DSKI$ command (SV)

This command is used to load information from a sector or sectors of a disk directly into the computer regardless of the file name or type. It takes the form -

DSKI$(a,b,c)

where a is the drive number, b is the track number and c is the sector number. For example

DSKI$(1,2,3) will read the third sector of the second track of the disk in drive 1.

The following program demonstrates how DSKI$ can be used to examine the sectors on a disk. It has been extracted from a larger program which is used to modify information stored on the disk.

```
10 CLEAR5000:DEFINTA-Z:DIMS$(16):FORI=0T
O15:READS$(I):NEXT
20 CLS:WIDTH40:LOCATE0,1:PRINT"ENTER DRI
VE NUMBER (1 OR 2)";:INPUTD:IFD<1ORD>2TH
EN20
30 LOCATE0,3:PRINT"STARTING TRACK (0-39)
";:INPUTST:IFST<0ORST>39THEN30
40 LOCATE0,5:PRINT"STARTING SECTOR ";:IF
ST=0THENPRINT"(1-18) ";ELSEPRINT"(1-17)
";
50 INPUTSS:IFSS<1ORSS>18THEN40
60 IFSS>128ANDST=0THEN40
70 CLS:FIELD#0,128ASA$,128ASB$:IFST<>0TH
EN170
80 L=1:I=0:FORJ=SSTO18:PRINT"          *** TR
ACK ";I;", SECTOR ";J;" ***":PRINT:GOSUB
220:PRINT
90 PRINT:PRINT"ENTER FOR NEXT, SPACE BAR
 TO SELECT NEW TRACK,SECTOR";
100 I$=INKEY$:IFI$=" "THENCLS:GOTO30ELSE
IFI$=CHR$(13)THENST=1:SS=1:GOTO120
110 GOTO100
120 CLS:NEXT
130 L=1:J=SS:FORI=STTO39
140 CLS:PRINT"      *** TRACK ";I;" SECTO
R ";J;" ***":PRINT:GOSUB220:PRINT
150 PRINT"ENTER FOR NEXT, SPACE BAR TO S
ELECT NEW TRACK,SECTOR";
160 I$=INKEY$:IFI$=" "THENCLS:GOTO30
170 IFI$=CHR$(13)THEN190
180 GOTO160
190 J=J+1:IFJ>17THENJ=1:GOTO210
200 GOTO140
210 CLS:NEXT
```

```
220 A$="":L=1:S=0:PRINTS$(S);" ";:A$=DSK
I$(D,I,J):Z=LEN(A$):FORX=1TOZ
230 B=ASC(MID$(A$,X,1)):IFB<16THENB$="0"
+HEX$(B)ELSEB$=HEX$(B)
240 IFB<16THENPRINT"0";HEX$(B);ELSEPRINT
HEX$(B);
250 L=L+1:IFL=17THENL=1:B=0:S=S+1:PRINT:
PRINTS$(S);" ";
260 NEXT:A$="":RETURN
270 DATA00,10,20,30,40,50,60,70,80,90,A0
,B0,C0,D0,E0,F0
```

## The DSKI$ command (MSX)

This command is similar to the one used for Spectravideo except that only two values are given with the command - drive number and logical sector number. The logical sector number is counted from zero using 9 sectors per track. Therefore the first sector of the third track is 18. The information is read into a block of memory the start address of which is stored in locations &HF351 and &HF352.

A sample program to read sector 18 would be -

```
10 CLEAR 1000
20 S$=DSKI$(1,18)
30 X=PEEK(&HF351)+256*PEEK(&HF352)
40 FOR I=X TO 256+X
50 PRINT HEX$(PEEK(X));" ";CHR$(PEEK(X))
60 NEXT
```

## The DSKO$ command (SV)

This command is the reverse of the DSKI$ command and is used to write information directly to a disk. It takes the form

```
DSKO$ a,b,c
```

where a is the drive number, b is the track number and c is the sector number.

## The DSKO$ command (MSX)

This command is the reverse of the DSKI$ command and is used to write information directly to a disk. It takes the form

```
DSKO$ a,b
```

where a is the drive number, b is the logical sector number starting at zero.

ANSWERS to the questions in issue 8

We do not normally supply answers to last issue's questions, but thought we might explain why some of those programs were wrong.

1. The command PRINT is spelt incorrectly - PRIMT

2. Line 110 should be READ A$ because the data is not numeric.

3. Nothing wrong

4. Nothing really wrong if you only want to read the first five data items. To read all eight change line 100 to read FOR I=1 TO 8

5. You cannot use graphics commands while in screen mode 0.

6. This program is in a 'dead loop' because the variable I will never equal 10 and line 130 loops back to line 100.

7. This program sets up screen 0 then jumps to line 140 and sets up screen 2 without doing anything else.

Want to try some more?

# QUESTIONS

1. What error message will this cause to be displayed?

```
100 A=10
110 GOSUB 500
120 PRINT A
500 A=A+2
510 RETURN
```

2. What error message will this cause to be displayed?

```
10 PRINT A(21)
```

3. What error message will this cause to be displayed?

```
100 GOTO 120
110 PRINT
130 END
```

# TUTORIAL INDEX (V1N3P13 = Volume 1, No 3, page 13)

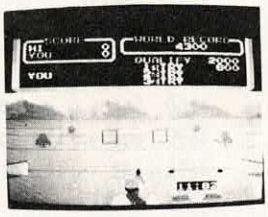

| | | | | | |
|---|---|---|---|---|---|
| Arrays | V1N3P13 | LINE INPUT | V1N4P12 | Searching | V1N9P14 |
| ASC | V1N7P13 | LIST | V1N1P12 | SET (disk) | V1N8P16 |
| ATTR$ (disk) | V1N8P15 | LOAD | V1N5P11 | Sorting | V1N9P11 |
| AUTO | V1N7P11 | LOCATE | V1N4P12 | SOUND | V1N3P19 |
| BEEP | V1N3P19 | Loops (nested) | V1N2P14 | | V1N4P19 |
| Binary notation | V1N4P16 | Manipulating strings | V1N3P11 | | V1N5P17 |
| BLOAD | V1N5P12 | Maths (simple) | V1N1P16 | SPACE$ | V1N3P13 |
| BSAVE | V1N5P12 | (computer) | V1N4P15 | Sprites | V1N6P17 |
| Cassette (load/save) | V1N5P9 | MAXFILES | V1N4P13 | | V1N7P18 |
| CHR$ | V1N7P14 | MERGE | V1N5P13 | collision | V1N8P12 |
| CIRCLE | V1N4P17 | MID$ | V1N3P12 | SPRITE$ | V1N6P18 |
| CLOAD | V1N5P10 | Missing Operand | V1N9P15 | STEP | V1N2P14 |
| CLOAD? | V1N5P14 | NAME (disk) | V1N8P14 | STICK | V1N8P9 |
| CLOSE | V1N4P13 | NEW | V1N1P12 | STOP | V1N1P17 |
| COLOR | V1N7P17 | NEXT | V1N2P13 | STOP ON/OFF | V1N9P9 |
| COPY (disk) | V1N8P15 | ON GOSUB | V1N2P11 | STRIG | V1N8P9 |
| CSAVE | V1N5P10 | ON GOTO | V1N2P11 | STRING MANIPULATION | V1N1P14 |
| DATA | V1N2P15 | ON KEY | V1N9P9 | | V1N3P11 |
| Decimal notation | V1N4P15 | ON SPRITE | V1N7P19 | | V1N3P13 |
| DELETE | V1N7P11 | ON STOP | V1N9P9 | + sign | V1N1P14 |
| DIM | V1N3P14 | ON STRIG | V1N8P10 | ; sign | V1N1P14 |
| Disk basic | V1N8P13 | OPEN | V1N4P13 | comma | V1N1P15 |
| | V1N9P16 | PAINT | V1N7P17 | STRING$ | V1N3P12 |
| DRAW | V1N5P16 | PEEK | V1N6P11 | STR$ | V1N7P14 |
| | V1N6P15 | PLAY | V1N4P20 | Subscript Out of Range | V1N9P16 |
| DSKF (disk) | V1N8P15 | | V1N5P18 | SWAP | V1N7P15 |
| DSKI$ | V1N9P17 | | V1N7P16 | Syntax Error | V1N9P15 |
| DSKO$ | V1N9P18 | POINT | V1N8P12 | THEN | V1N2P17 |
| ELSE | V1N2P18 | POKE | V1N6P13 | TROFF | V1N7P12 |
| Error messages | V1N9P15 | PRESET | V1N8P11 | TRON | V1N7P12 |
| FILES (disk) | V1N8P13 | PRINT | V1N1P11 | Type Mismatch | V1N9P15 |
| FOR | V1N2P13 | PRINT# | V1N4P14 | VAL | V1N7P14 |
| FRE | V1N7P16 | Programs (entering) | V1N7P9 | Variables | V1N1P13 |
| GOSUB | V1N2P10 | De-bugging | V1N8P16 | | V1N7P13 |
| GOTO | V1N1P18 | Programming technique | V1N7P10 | | V1N8P8 |
| Hexadecimal notation | V1N4P17 | | V1N9P11 | VPEEK | V1N6P14 |
| IF | V1N2P17 | PSET | V1N8P11 | VPOKE | V1N6P14 |
| Illegal Function Call | V1N9P16 | PUT SPRITE | V1N6P21 | | |
| INKEY$ | V1N4P11 | | V1N7P18 | | |
| INPUT | V1N1P17 | Random numbers | V1N8P7 | | |
| | V1N2P9 | READ | V1N2P15 | | |
| INPUT$ | V1N4P12 | Relational operators | V1N2P18 | | |
| INPUT# | V1N4P14 | REM | V1N2P13 | | |
| INSTR | V1N7P15 | RENUM | V1N7P11 | | |
| Joystick | V1N8P9 | RESTORE | V1N2P16 | | |
| KEY | V1N7P12 | RETURN | V1N2P10 | | |
| KEY LIST | V1N9P10 | RIGHT$ | V1N3P11 | | |
| KEY () ON | V1N9P10 | RND | V1N8P7 | | |
| KILL (disk) | V1N8P14 | RUN | V1N1P12 | | |
| LEFT$ | V1N3P11 | SAVE | V1N5P11 | | |
| LEN | V1N3P12 | SCREEN | V1N3P17 | | |
| LFILES (disk) | V1N8P14 | Saving (MSX) | V1N6P11 | | |
| LINE | V1N5P15 | Saving (SVI) | V1N5P10 | | |

Charlie does his stuff on the circus favourite, the flying trapeze. All this isn't easy for Circus Charlie, but, as the saying goes - "Practice Makes Perfect!"

1 or 2 player, good music and graphics - addictive!

Publisher: Konami
Price: $39.95

## HYPER SPORTS II (MSX Cartridge)

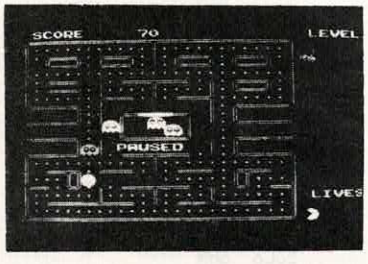


You are a competitor in three sports events. In the skeet shoot the gun is aimed for you and you must hit as many targets as possible. If you exceed the qualifying score, you move on to archery. Wind strength and direction are randomly chosen. The target moves across your line of fire and the length of time that the trigger is held determines the angle of flight and therefore height on the target. If you qualify in archery you can move on to weightlifting. This one will kill your joystick. You select the weight and then franticly waggle the joystick to lift it. When the weight is at chest height, pressing the trigger will snap it to overhead and you continue waggling until you get three lights.

Requires great concentration and practice. 1 player, good music and graphics - strenuous!

Publisher: Konami
Price: $49.95

## OH NO! (32K MSX Cassette)

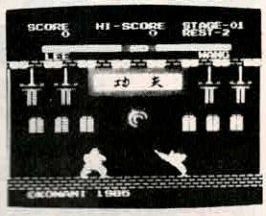



Based on the popular Pac-man style, OH NO! uses a synthetic voice to introduce the ghosts - Joey, Paul, Willie and Frankie. The idea is to move around the maze collecting food and avoiding the ghosts. If you eat a power pill then you are able to eat the ghosts for a while. Fruit also appears on the screen to be eaten for bonus points.

There is a story around that this was originally called OH (unprintable)! and that was what the main character spoke when he died. In our version, the same synthetic voice just said OH NO! GAME OVER.

Publisher: Aackosoft
Price: $14.95

## YIE AR KUNG-FU (MSX Cartridge)

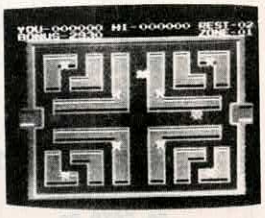


Kung-fu fighter, Lee, has to defeat each opponent he faces by delivering nine punches or kicks. At the top of the screen are the energy levels of you and your opponent. These diminish as either of you sustain a hit. Between every two opponents is a bonus stage where you must destroy such objects as bricks and vases before they hit you. You receive an extra life at 30000 and 50000.

One player, good music and graphics. Vicious!

Publisher: Konami
Price: $49.95

## MOPIRANGER (MSX Cartridge)

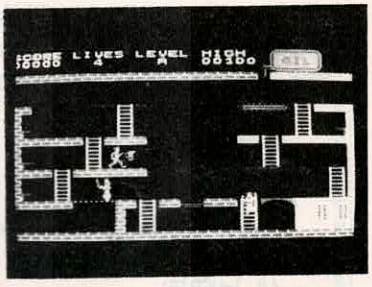


Mopi Ranger, in his little boat, must row around the obstacles to rescue all the Moplits. He also must watch out for the Razons; one touch from them and Mopi Ranger's boat is sunk. He can shoot at them and turn them to stone blocks which he can move one at a time. Beware of the big Razon, though, because you can't shoot him! You must remember not to box yourself in.

1 player, very good music, average graphics - great!

Publisher: Konami
Price: $39.95

## TURMOIL (64K MSX Cassette)



Mic the mechanic has been taken prisoner by an Arab oil Sheik. To escape he must collect oil from a dripping drum in order to build a get-away car. He can collect tools which appear around the place to gain more points. All the while, Mic is being pursued by athletic Arabs brandishing swords. He must avoid them and collect the oil and tools in this 26 screen game.

1 player, good graphics - difficult.

Publisher: Bug-Byte
Price: $12.95

# HACKERS CORNER

We received a request from Tim Jessel of Victoria for assistance with a machine language routine he had used to set a pixel in the centre of the graphics screen on his Spectravideo. Although we are not in the habit of de-bugging readers' programs, (and we don't intend to start), we thought we would use this as a demonstration of using the inbuilt ROM routines from a machine code program.

This is our corrected version of his routine -

```
10 REM &H4980 pixel colour in A register
11 REM &H48E9 position graphics cursor
12 REM at x,y - x in BC and y in DE
13 REM &H4988 sets pixel at x,y to the
14 REM colour in register A
20 CLEAR 200,&HD000
30 DEFUSR=&HD000
40 FOR I=0 TO 17 : READ A$
50 POKE &HD000+I,VAL("&H"+A$) : NEXT
60 SCREEN 1 : REM this could be done in M/L
70 J=USR(0)
80 GOTO 80
90 REM Get colour (15) into A register
91 DATA 3E,0F    : REM LD A,0FH
92 DATA CD,80,49 : REM CALL &H4980
93 REM x=128 (decimal) into BC
94 DATA 01,80,00 : REM LD BC,80H
95 REM y=96 (decimal) into DE
96 DATA 11,60,00 : REM LD DE,60H
97 DATA CD,E9,48 : REM CALL &H48E9
98 DATA CD,88,49 : REM CALL &H4988
99 DATA C9       : REM RETURN
```

MSX version calls MAPXYC at 0111H and SETC at 0120H. Note that all registers must be saved (line 80) and then restored afterwards (line 85).

```
10 CLEAR 200,&HD000
20 DEFUSR=&HD000
30 SCREEN 2
40 FOR I=0 TO 20 : READ A$
50 POKE &HD000+I,VAL("&H"+A$) : NEXT
60 J=USR(0)
70 GOTO 70
80 DATA C5,D5,E5,F5
81 DATA 01,80,00
82 DATA 11,60,00
83 DATA CD,11,01
84 DATA CD,20,01
85 DATA F1,E1,D1,C1
86 DATA C9
```

## THE SV ROM DECODED

Spectravideo SV-318/328 ROM version 1.1

```
4552  COLOR
459A  SCREEN
4606  SPRITE
4623  SPRITE$
464C  PUT SPRITE
466D  Get x,y of sprite
469E  Get sprite color
46BE  Get sprite plane
46D8  VPOKE
46F2  VPEEK
4782  Spectravideo Logo
481A  SETBIGPOINT
48E1  Check screen mode
48E9  Loc GRP cursor (x in BC, y in DE)
4951  Read colour of x,y pixel into A
4980  Send colour in A to FA13H
4988  Set pixel to colour in FA13H
49CF  GRP cursor right one pixel
49F8  GRP cursor left one pixel
4A14  GRP cursor down, check
      for bottom line
4A2D  GRP cursor down
4A3F  GRP cursor up, check for top line
4A59  GRP cursor up
4AF9  Draw line at BC,DE for length L
50B8  COS
50D1  SIN
5120  TAN
5139  ATN
5197  LOG
5222  SQR
526B  EXP
5300  RND
5355  Multiply by 2
5361 se  power of 2
55B1  ABS
55C6  SGN
56B5  CINT
56DD  CSNG
5765  CDBL
57E9  FIX
57F8  INT
59CB  ASCII to binary
5B44  HEX to DEC
6061  DIM
6066  Get variable HL=var name ptr
      DE=variable address ptr
```

## MSX ROM CALLS

009CH Name: CHSNS
Function: Check keybd buffer status
Entry: None
Returns: Reset Z flag if a char in buffer
Modifies: AF

009FH Name: CHGET
Function: Waits for input and returns the character
Entry: None
Returns: Character code in A
Modifies: AF

00A2H Name: CHPUT
Function: Display a character
Entry: Character code in A
Returns: None
Modifies: None

00A5H Name: LPTOUT
Function: Send character to printer
Entry: Character code in A
Returns: Abort sets carry flag
Modifies: F

00A8H Name: LPTSTT
Function: Check printer status
Entry: None
Returns: Ready=255 in A & Z flag reset
Busy =0 in A & Z flag set
Modifies: AF

00ABH Name: CNVCHR
Function: Check for graphics, change code
Entry: Char code in A
Returns: Carry reset=graphic header byte
Carry & Z set=converted graphic
Carry set, Z reset = non-converted code
Modifies: AF

00AEH Name: PINLIN
Function: Accept input line up to CR or STOP and store in line buffer
Entry: None
Returns: Top of buffer -1 in HL, carry set if STOP entered
Modifies: All

00B1H Name: INLIN
Function: Same as PINLIN
Entry: Same as PINLIN
Returns: Same as PINLIN
Modifies: All

# BOOKS • BOOKS

## SIMPLE INTERFACING PROJECTS by Owen Bishop




Although this book is definitely not for the novice, it provides some background information to the subject of digital electronics as they relate to the computer.

The book is based on 12 projects to build for your computer. All of these can be used with SV and MSX computers with little or no modification. The projects include a real-time clock, graphics tablet, ROM board and programmer, acoustic coupled modem and a digitiser pad. Each is described together with diagrams and parts list.

The introduction explains interfacing techniques, address decoding, reading and writing, connecting to busses, logic levels, power supplies, decoupling and construction methods.

For the more experienced there is useful information such as calculating the propogation delay in a circuit. This is a handy book if only to learn how a computer controls external devices.

Review copy supplied by Peter J Fox Computers

Publisher: Granada Publishing
168 pages
Price $5.00




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We are the Australian agents for Panda Soft International. Please write for a full price list or further information.

# CLUB NEWS

My name is Erik Kowarski. I am a musician in Perth, a CX5M owner and member of the MSX Users Group of WA. I have received several questions about these machines. Unfortunately one reader's letter was in my bag which was recently stolen so I cannot reply directly. Since I cannot remember the writer's name I thought this article would be the best solution.

The main question seems to relate to memory size. All MSX computers give the same free memory when powering up, yet some software, mostly machine code adventure games will not work on the CX5.

Well, as was mentioned a couple of issues ago, the MSX standard does not specify the size of user programmable memory or RAM, and whereas most MSX machines have 64 kilobytes of RAM, the Yamaha CX5M has only 32 kilobytes.

The Z80 processor, the brain of all MSX computers, has 16 bit addressing, that is, it can address or talk to no more than 64K of memory at one time.

Most of us write our own programs in Basic. This language is not natural to the Z80, so it carries an interpreter around to understand our programs. The MSX interpreter is quite fancy - that is why we have a full screen editor, sprite graphics, and over 150 Basic commands. It also takes up 32K of memory area stored in ROM - Read Only Memory not available to the user. This only leaves 32K of RAM which the computer can address for use in Basic programs.

When we switch on an MSX machine, the screen tells us we have 28815 bytes free (disk users have a little less). Yet I have just said we have 32K (32768 bytes) of RAM available in Basic - so what has happened to nearly 4000 bytes? The computer needs these to store the information used by the Basic interpreter.

Now let us talk about machine code. This is the language the computer does understand without an interpreter. It is the computer's own internal language. If a program is written in machine code then almost the full 64K of addressable memory can be used and not just 28815 bytes.

Most MSX computers have a full 64K of RAM for use in machine code programs as well as the 32K Basic in ROM, but the Yamaha CX5M only has 32K RAM. This is why some machine code programs will not fit the CX5 and therefore will not run.

I hope this makes the situation a little clearer. Next issue I would like to print a list of software known to need more than 28K to run, so if anyone has had any such experience, please write in and let us know.

Cheers, Erik.

# USER GROUPS

Adelaide area should contact
JOE ROWLEY
255 Hogarth Road, Elizabeth Grove
South Australia 5112. Phone 255 6738

## SYDNEY USERS

We have been advised of a group in Sydney for Spectravideo owners– please contact Malcolm Samuels on 888 7664 after hours.

## QUEENSLAND

BRISBANE SPECTRAVIDEO & MSX USERS GROUP
C/- Mrs Lucille Parker (Secretary)
25 Primrose Street, Woodridge, Queensland 4114

## VICTORIA

MELBOURNE SPECTRAVIDEO & MSX USERS GROUP
C/- John Halford, 48 Shawlands Avenue
Blackburn South, Victoria 3130
Phone (03) 878 6169

GIPPSLAND MSX USER GROUP
C/- George Francis, 31 Donald Street
Morwell, Victoria 3840
Phone 051-34 5175

## WESTERN AUSTRALIA

THE MSX USERS GROUP
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Mt Hawthorn, W.A. 6016

## NEW ZEALAND

WELLINGTON SPECTRAVIDEO USERS GROUP
P.O. Box 26-050
Newlands, Wellington 6004, New Zealand

## CANADA

SMUG of Montreal
20 - 159th Avenue
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## FINLAND

SVI-MSX Club/Turku
C/- Mika Ristolainen
Lumikonkatu 7H 272
SF-20240 Turku
FINLAND

# COMPUTER KIDS

Hi Kids! After the last Computer Kids, I received a great letter and a neat little program from Miss Lee Bergman. She hopes that all you other Computer Kids readers will get a lot of fun from it. Thanks, Lee! It's good to see that kids these days can still do things for other people.

Why don't some of you other readers take the trouble to write a little program to send in or handy tips to other junior programmers. Just think, you'll be helping other kids ... and you'll have your name in a national magazine to show your friends and family!

Oh well, it's time to go again. I hope all you readers have a great time typing in Lee's ace program, and I'll sign off until the next issue.

Bye for now, Wendy.

## YACHT RACE (MSX)
### Written by Lee Bergman

```
10 OPEN "GRP:" AS 1
15 PLAY"T255o4CDET255o4CDET255o4CDE"
20 SCREEN 3,3
30 PRINT#1," YACHT"
40 PRINT#1,"  RACE"
50 FOR I=1 TO 2000:NEXT I:CLS
60 SCREEN2,2:PRINT#1,"    by Lee Bergman
"
70 PRINT#1,"                December 1986"
80 FOR D=1 TO 2000:NEXT D:CLS
90 COLOR3,5,5
100 PRINT#1,"      ■■"
110 PRINT#1,"      ■■■"
120 PRINT#1,"      ■■■■"
130 PRINT#1,"      ■■■■■"
140 PRINT#1,"      ■"
150 PRINT#1,"    ■■■■■■■"
160 PRINT#1,"     ■■■■■"
170 PRINT#1,"      ■■■■"
180 FOR F=1 TO 2000:NEXT F:CLS
190 COLOR15,4,4
NOTE: The blocks are Graph key + P
```

## READERS ADS

Use these pages to sell your surplus equipment, etc. Photocopy this form and insert up to 20 words. Put it in an envelope with $5 and send to COMPUTER FORUM by the 10th of each month.

| | | | |
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## PROGRAM LISTINGS

**Each issue of COMPUTER FORUM will contain program listings for you to type in. When entering these programs it is a good idea to save off to cassette or disk every 10 or 20 lines and then continue. Then if the power goes off or another major disaster occurs, you will not have to type in the whole lot again.**

**Another check is to verify each save using the CSAVE? command. To use this command, CSAVE your program, then rewind the tape and type CLOAD? and press enter. This command will then compare the program on the tape with the one in memory, to verify that it was saved correctly.**

**All of the listings are printed in 40 column format. If you enter WIDTH40 before typing them in, you will be able to line up the characters in the listing with the text on screen.**

**Take care that you do not mistake the letter l for 1 or an O for a zero. Where graphics characters are used in the program, they will be shown as @1@. This symbol will then be repeated at the end of the listing with instructions concerning the graphics character used.**

**All of the listings have been tested on MSX and SV318/328 computers and do work. Should you have any problems, we suggest that you thoroughly check your work as errors can creep in very easily. Should you not wish to type in these listings, send a blank tape together with $2 to COMPUTER FORUM and we will post a copy to you. Do not forget to tell us which computer you have.**

# BIKER

by S. Apps

A graphic motorcycle jump program.

This listing is in Spectravideo SV318/328 format. For MSX owners, see the note at the end of the listing.

```
1 REM BIKER by S. APPS
2 REM REQUIRES CAREFUL USE OF THE JOY-
3 REM STICK TO MAKE THE BIKER MOVE AND
4 REM JUMP.  LEFT ON THE STICK MAKES
5 REM HIM MOVE, LEFT  AND UP RAISES THE
6 REM FRONT WHEEL, UP MAKES HIM JUMP.
10 COLOR 15,4,5:SCREEN1,2
20 DEFINTA-Z:LE=1
30 FOR S=1 TO 5:A$="":FOR I=1 TO 32:READ
 A:A$=A$+CHR$(A):NEXT I:SPRITE$(S)=A$:NE
XT S
40 FOR I=1 TO 8 :READ A:F$=F$+CHR$(A):NE
XT:SPRITE$(6)=F$
50 BEEP:LINE(60,0)-(210,10),6,BF
60 COLOR 15,4,LE
70 LOCATE80,0:PRINT"SCORE";SC:LOCATE130,
0:PRINT"   LEVEL";LE
90 LINE(0,160)-(300,180),LE+4,BF
90 LINE(100,160)-(140,180),4,BF
100 LINE(0,100)-(300,120),LE+4 ,BF
110 LINE(80,100)-(110,120),4,BF
120 LINE(0,40)-(300,60),LE+4 ,BF
130 LINE(150,40)-(190,60),4,BF
140 Q=23:A=0:X=230:Y=143:PUT SPRITE1,(X,
Y),1,1:SPRITEON
150 D=STICK(0) OR STICK(1)
160 ON SPRITE GOSUB 650
170 W=W+F+2
180 IF W+Z>190 THEN W=-30
190 IF V>=1 THEN PUT SPRITE3,(Z,Q),2,3
200 IF V>=2 THEN PUT SPRITE4,(Z+30,Q+60)
,15,3
210 IF V>=3 AND W>0 THEN PUT SPRITE5,(Z+
W+25,Q-3),15,4
220 IF V>=4 AND W>0 THEN PUT SPRITE6,(Z+
W+65,Q+57),2,4
230 IF V>=5 THEN PUT SPRITE7,(90,60),7,6
240 IF V>=6 THEN PUT SPRITE8,(120,130),8
,6
250 IF V>=7 THEN PUT SPRITE9,(170,10),9,
6
260 IF V>=8 THEN PUT SPRITE10,(20,120),5
,6
270 IF V>=9 THEN F=3
280 IF V=10 THEN F=4
290 IF Q>143 THEN Q=Q-180
300 IF V>=1 THEN Z=Z+F+1:IF Z=200 THEN Z
=0:Q=Q+60
310 IF LE>10 THEN CLS:PRINT "  you are t
he best"
320 IF X<0 AND Y<50 AND A=0 THEN V=V+1:S
C=SC+50:LE=LE+1:CLS:GOTO 50
```

```
330 IF X<0 THEN X=240:Y=Y-60
340 IF A>0 THEN 360
350 IF POINT(X+8,Y+20)=4 THEN GOTO 650
360 IF D=1 OR D=8 THEN PUT SPRITE1,(X,Y)
,1,2:GOTO 380
370 PUT SPRITE1,(X,Y),1,1
380 IF D=7 AND A=0 THEN X=X-5:C=0
390 IF D=0 AND A>0 THEN X=X-6:Y=Y+4:A=A-
1:B=1:C=0
400 IF D=1 AND B=1 AND A>0 THEN GOTO 650
410 IF D=1 AND C=1 THEN X=X-6:Y=Y-4:A=A+
1:B=0
420 IF D=8 AND A=0 THEN X=X-8:C=1
430 GOTO 150
440 DATA 0,0,0,0,0,1,2,3
450 DATA 3,5,57,89,181,180,72,48
460 DATA 12,4,12,6,254,6,62,207
470 DATA 223,254,252,250,173,173,18,12
480 DATA 0,0,0,0,0,0,24,39
490 DATA 94,90,37,24,0,0,0,0
500 DATA 0,0,3,1,115,137,255,94
510 DATA 252,255,238,94,45,45,18,12
520 DATA 7,15,15,7,2,7,15,23
530 DATA 47,95,127,127,175,87,32,0
540 DATA 0,224,224,0,0,0,128,64
550 DATA 160,252,254,254,235,212,8,0
560 DATA 0,0,16,24,156,63,127,63
570 DATA 128,0,0,0,0,0,0,0
580 DATA 0,0,0,64,96,252,254,255
590 DATA 96,96,64,0,0,0,0,0
600 DATA 2,57,0 ,50,222,0 ,30,249
610 DATA 3,87,45,0 ,123,234,4,0
620 DATA 0,56,87,0,32,0,222,0
630 DATA 88,66,33,0,0,11,55,0
640 DATA 255,255,255,255,255,255,255,255
650 SPRITE OFF:PUT SPRITE1,(X,Y),1,5:SC=
SC-10:SOUND0,0:SOUND6,10:SOUND7,20:SOUND
8,16:SOUND12,30:SOUND13,0:FOR I=1 TO 100
0:NEXT:CLS
660 GOTO 50
```

**NOTE:** FOR MSX CHANGE THE FOLLOWING LINES TO READ -

```
10 COLOR 15,4,5:SCREEN2,2:OPEN"GRP:"AS1
70 PRESET(76,2):PRINT#1,"SCORE";SC:PRESE
T(126,2):PRINT#1,"   LEVEL";LE
310 IF LE>10 THEN CLS:PRINT#1,"  you are
 the best"
660 PUT SPRITE 1,(0,209):PUT SPRITE 2,(0
,209):GOTO 50
```

# SPRITE DEFINER
## by Erik Kowarski

A useful program to help you design sprites.

MSX FORMAT ONLY

```
10 ' First select 8*8 or 16*16- ---input
 can be by keyboard..(wait 5 seconds) or
 joystick or cursor keys.          Colour
 squares on grid by coordinates (keyboar
d- use no comma) or fire/space bar.
20 ' To see sprite, press F1-left & righ
t stick give magnified or normal display
... Tomodify image, press F3. When happy
, print text to screen by  pressing F2
-this puts you in direct mode.
30 ' Press F6 to erase program, then  ad
d line numbers to the text (don't  forge
t to replace"X"with your pattern no.).
40 ' To save text to disk, press F7,   &
 then add rest of filename+",A" for late
r mergeing....F8 loads sprite makerback
 in ready for new image.
50 KEYOFF:CLEAR500:DEFINTD,H,R,S,W:W=0
60 SCREEN1,0:COLOR15,4,4:CLS:KEY1,"--":K
EY2,"pp":KEY3,"":KEY5,"RUN"+CHR$(13):KEY
6,"DELETE-7000"+CHR$(13):KEY7,"SAVE"+CHR
$(34)+"SP":KEY8,"RUN"+CHR$(34)+"SPRITE"+
CHR$(34)+CHR$(13)
70 SPRITE$(0)=CHR$(0)+CHR$(0)+CHR$(24)+C
HR$(60)+CHR$(60)+CHR$(24)+CHR$(0)+CHR$(0
)
80 LOCATE5,1:PRINT"___SPRITE MAKER____":
PRINT:PRINT"BY ERIK KOWARSKI      25/4/8
6"
90 LOCATE2,6:PRINT"F 1=see sprite(sp sp)
"+SPACE$(8)+"< >=change sprite size"+SPA
CE$(7)+"F 2=printout(P P)"+SPACE$(12)+"F
 3=edit-*USE F3 AFTER F1!!"+SPACE$(2)+"F
 5=BACK TO MENU"
100 LOCATE2,12:PRINT"F 6=ERASE PROGRAM"+
SPACE$(12)+"F 7=SAVE YOUR SPRITE"+SPACE$
(9)+"F 8=LOAD-RUN"
110 LOCATE4,17:PRINT"SELECT  1= 8 * 8":L
OCATE12,18:PRINT"2= 16*16":LOCATE12,19:P
RINT"3= DEMO- < >"
120 LOCATE0,22:PRINT"--    pp    EDIT  1
ist  RUN"
130 LOCATE2,17:A$=INPUT$(1):IFA$="1"GOTO
140ELSEIFA$="2"GOTO500ELSEIFA$="3"GOTO17
10ELSEGOTO130
140 '-----8*8-----
150 CLEAR2000
```

```
160 H=0:R=64:S=64:DIMQ$(7),Z$(16,16),Q(7
)
170 SCREEN2:OPEN"GRP:"FOROUTPUTAS#1
180 GOSUB920
190 TT$="01234567":PRESET(66,48):PRINT#1
,TT$:PRESET(66,144):PRINT#1,TT$
200 FORI=0TO7:PRESET(50,66+8*I):PRINT#1,
I;SPACE$(6);I;"  ";
210 IFI=3THENPRINT#1,"F1=SEE"ELSEIFI=5TH
ENPRINT#1,"F2=PRNT"ELSEIFI=7THENPRINT#1,
"F3=EDIT"
220 NEXT:PRESET(50,177):PRINT#1,"INPUT X
Y":CLOSE#1
230 FORI=0TO8:LINE(I*8+64,64)-STEP(0,64)
:LINE(64,I*8+64)-STEP(64,0):NEXT
240 FORJ=0TO7:FORI=0TO7:Z$(I,J)="0":NEXT
:NEXT
250 IFP=10THENGOSUB1410:GOTO1010ELSEGOSU
B1460
260 C$=INPUT$(2)
270 IFC$="  "ORC$="--"GOTO390
280 IFC$="PP"ORC$="pp"THENGOSUB460
290 A$=LEFT$(C$,1):B$=RIGHT$(C$,1)
300 IFA$<"0"ORA$>"7"GOTO260
310 IFB$<"0"ORB$>"7"GOTO260
320 OPEN"GRP:"FOROUTPUTAS#1
330 LINE(98,176)-STEP(24,8),4,BF
340 PRESET(98,176):PRINT#1,LEFT$(C$,1);"
,";RIGHT$(C$,1):CLOSE#1
350 A=VAL(A$):B=VAL(B$)
360 IFZ$(A,B)="1"THENLINE(A*8+65,B*8+65)
-STEP(6,6),4,BF:Z$(A,B)="0":GOTO260
370 LINE(A*8+65,B*8+65)-STEP(6,6),10,BF
380 Z$(A,B)="1":GOTO260
390 '-----EVALUATE-----
400 FORJ=0TO7:Q$(J)="":NEXT
410 FORJ=0TO7:FORI=0TO7
420 Q$(J)=Q$(J)+Z$(I,J):NEXT:NEXT
430 FORJ=0TO7:Q(J)=VAL("&B"+Q$(J)):NEXT
440 FORJ=0TO7:Q$(J)=STR$(Q(J)):NEXT:IFH=
1THENRETURN
450 GOTO1320
460 '-----PRINTOUT-----
470 KEY(2)OFF:H=1:SCREEN0:KEYON:GOSUB390
:LOCATE0,4
480 PRINT"     SPRITE$(X)=CHR$("+Q$(0)+"
)+CHR$("+Q$(1)+")+CHR$("+Q$(2)+")+CHR$("
+Q$(3)+")+CHR$("+Q$(4)+")+CHR$("+Q$(5)+"
)+CHR$("+Q$(6)+")+CHR$("+Q$(7)+")"
490 RETURN910
500 '-----16*16-----
510 CLEAR8000:H=0:R=56:S=20:DIMQ$(31),Z$
(16,16),Q(31)
520 SCREEN2:OPEN"GRP:"FOROUTPUTAS#1
530 GOSUB920
540 TT$="0123456789ABCDEF":PRESET(58,4):
PRINT#1,TT$:PRESET(58,157):PRINT#1,TT$
550 FORI=1TO16
560 PRESET(42,13+8*I):PRINT#1,MID$(TT$,I
,1)+SPACE$(18)+MID$(TT$,I,1)+"  ";
570 IFI=9THENPRINT#1,"F1="ELSEIFI=10THEN
PRINT#1,"SEE"ELSEIFI=12THENPRINT#1,"F2="
ELSEIFI=13THENPRINT#1,"PRT"
580 IFI=15THENPRINT#1,"F3="ELSEIFI=16THE
NPRINT#1,"ED."
590 NEXT:PRESET(42,177):PRINT#1,"INPUT X
Y":CLOSE#1
600 FORI=0TO16:LINE(I*8+56,20)-STEP(0,12
8):LINE(56,I*8+20)-STEP(128,0):NEXT
610 FORJ=0TO15:FORI=0TO15:Z$(I,J)="0":NE
XT:NEXT
620 IFP=10THENGOSUB1620:GOTO1010ELSEGOSU
B1670
630 C$=INPUT$(2)
640 IFC$="  "ORC$="--"GOTO770
650 IFC$="PP"ORC$="pp"THENGOSUB830
660 A$=LEFT$(C$,1):B$=RIGHT$(C$,1)
670 IFA$<"0"ORA$>"f"ORA$>"F"ANDA$<"a"GOT
0630
680 IFB$<"0"ORB$>"f"ORB$>"F"ANDB$<"a"GOT
0630
690 OPEN"GRP:"FOROUTPUTAS#1
700 LINE(88,177)-STEP(24,8),4,BF
710 PRESET(88,177):PRINT#1,A$;",";B$
720 CLOSE#1:A=VAL(A$):B=VAL(B$)
730 IFA$="A"ORA$="a"THENA=10ELSEIFA$="B"
ORA$="b"THENA=11ELSEIFA$="C"ORA$="c"THEN
A=12ELSEIFA$="D"ORA$="d"THENA=13ELSEIFA$
="E"ORA$="e"THENA=14ELSEIFA$="F"ORA$="f"
THENA=15
740 IFB$="A"ORB$="a"THENB=10ELSEIFB$="B"
ORB$="b"THENB=11ELSEIFB$="C"ORB$="c"THEN
B=12ELSEIFB$="D"ORB$="d"THENB=13ELSEIFB$
="E"ORB$="e"THENB=14ELSEIFB$="F"ORB$="f"
THENB=15
750 IFZ$(A,B)="1"THENLINE(A*8+57,B*8+21)
-STEP(6,6),4,BF:Z$(A,B)="0":GOTO630
760 LINE(A*8+57,B*8+21)-STEP(6,6),10,BF:
Z$(A,B)="1":GOTO630
770 '-----EVALUATE-----
780 FORJ=0TO31:Q$(J)="":NEXT
790 FORJ=0TO15:FORI=0TO7:Q$(J)=Q$(J)+Z$(
I,J):NEXT:NEXT
800 FORJ=0TO15:FORI=8TO15:Q$(J+16)=Q$(J+
16)+Z$(I,J):NEXT:NEXT
810 FORJ=0TO31:Q(J)=VAL("&B"+Q$(J)):Q$(J
)=STR$(Q(J)):NEXT:IFH=1THENRETURN
820 GOTO1500
830 '-----PRINTOUT-----
840 KEY(2)OFF:H=1:SCREEN0:KEYON:GOSUB770
:LOCATE0,2
850 PRINT"     V$=CHR$("+Q$(0)+")+CHR$("
+Q$(1)+")+CHR$("+Q$(2)+")+CHR$("+Q$(3)+"
)+CHR$("+Q$(4)+")+CHR$("+Q$(5)+")+CHR$("
+Q$(6)+")+CHR$("+Q$(7)+")"
860 PRINT"     W$=CHR$("+Q$(8)+")+CHR$("
+Q$(9)+")+CHR$("+Q$(10)+")+CHR$("+Q$(11)
+")+CHR$("+Q$(12)+")+CHR$("+Q$(13)+")+CH
R$("+Q$(14)+")+CHR$("+Q$(15)+")"
870 PRINT"     X$=CHR$("+Q$(16)+")+CHR$(
"+Q$(17)+")+CHR$("+Q$(18)+")+CHR$("+Q$(1
9)+")+CHR$("+Q$(20)+")+CHR$("+Q$(21)+")+
CHR$("+Q$(22)+")+CHR$("+Q$(23)+")"
```

```
880 PRINT"      Y$=CHR$("+Q$(24)+")+CHR$(
"+Q$(25)+")+CHR$("+Q$(26)+")+CHR$("+Q$(2
7)+")+CHR$("+Q$(28)+")+CHR$("+Q$(29)+")+
CHR$("+Q$(30)+")+CHR$("+Q$(31)+")"
890 PRINT"      SPRITE$(X)=V$+W$+X$+Y$"
900 RETURN910
910 END
920 '--SELECT STICK0,1OR2--
930 P=1:PRESET(64,90)
940 PRINT#1,"SELECT JOYSTICK"+SPACE$(47)
+"FIRE 1=SWITCH BETWEEN"+SPACE$(11)+"COL
OR AND TRANSPARENT"
950 ONSTRIGGOSUB1250,1260,1270
960 STRIG(0)ON:STRIG(1)ON:STRIG(2)ON
970 FORI=1TO3000:NEXT
980 LINE(48,90)-STEP(192,32),4,BF
990 STRIG(0)OFF:STRIG(1)OFF:STRIG(2)OFF
1000 RETURN
1010 '--MOVE STICK CURSOR--
1020 LINE(42,177)-STEP(80,8),4,BF
1030 A=0:B=0
1040 D=STICK(W):IFD=1THENB=B-1
1050 IFD=2THENA=A+1:B=B-1
1060 IFD=3THENA=A+1
1070 IFD=4THENA=A+1:B=B+1
1080 IFD=5THENB=B+1
1090 IFD=6THENA=A-1:B=B+1
1100 IFD=7THENA=A-1
1110 IFD=8THENA=A-1:B=B-1
1120 IFA<0THENA=0
1130 IFB<0THENB=0
1140 IFR=64ANDA>7THENA=7ELSEGOTO1150
1150 IFR=64ANDB>7THENB=7ELSEGOTO1160
1160 IFA>15THENA=15
1170 IFB>15THENB=15
1180 PUTSPRITE0,(A*8+R,B*8+S),9,0:FORI=1
TO100:NEXT
1190 ONSTRIGGOSUB1210,1210,1210
1200 STRIG(W)ON:GOTO1040
1210 '---FIRE1 (CHANGE COLOR, UPDATE Z$(
A,B))---
1220 STRIG(W)OFF
1230 IFZ$(A,B)="1"THENLINE(A*8+R+1,B*8+S
+1)-STEP(6,6),4,BF:Z$(A,B)="0":RETURN104
0
1240 LINE(A*8+R+1,B*8+S+1)-STEP(6,6),10,
BF:Z$(A,B)="1":RETURN1040
1250 W=0:P=10:RETURN980
1260 W=1:P=10:RETURN980
1270 W=2:P=10:RETURN980
1280 '---EDIT---
1290 SCREEN,0
1300 SPRITE$(0)=CHR$(0)+CHR$(0)+CHR$(24)
+CHR$(60)+CHR$(60)+CHR$(24)+CHR$(0)+CHR$
(0)
1310 RETURN1010
1320 '---DRAW 8*8 SPRITE---
1330 M=0:SCREEN,0
1340 SPRITE$(1)=CHR$(Q(0))+CHR$(Q(1))+CH
R$(Q(2))+CHR$(Q(3))+CHR$(Q(4))+CHR$(Q(5)
)+CHR$(Q(6))+CHR$(Q(7))
1350 PUTSPRITE1,(162,48),15,1
1360 D=STICK(W)
1370 IFD=0GOTO1360
1380 IFD=3THENM=1:SCREEN,1:GOTO1340
1390 IFD=7THENM=0:GOTO1320
1400 GOTO1400
1410 '---F1=SPRITE,F2=PRINT,F3=EDIT---
1420 ONKEYGOSUB1450,460,1280
1430 KEY(1)ON:KEY(2)ON:KEY(3)ON
1440 RETURN
1450 RETURN390
1460 ONKEYGOSUB,,1490
1470 KEY(3)ON
1480 RETURN
1490 RETURN260
1500 '---DRAW 16*16 SPRITE---
1510 M=0:SCREEN,2
1520 V$=CHR$(Q(0))+CHR$(Q(1))+CHR$(Q(2))
+CHR$(Q(3))+CHR$(Q(4))+CHR$(Q(5))+CHR$(Q
(6))+CHR$(Q(7))
1530 W$=CHR$(Q(8))+CHR$(Q(9))+CHR$(Q(10)
)+CHR$(Q(11))+CHR$(Q(12))+CHR$(Q(13))+CH
R$(Q(14))+CHR$(Q(15))
1540 X$=CHR$(Q(16))+CHR$(Q(17))+CHR$(Q(1
8))+CHR$(Q(19))+CHR$(Q(20))+CHR$(Q(21))+
CHR$(Q(22))+CHR$(Q(23))
1550 Y$=CHR$(Q(24))+CHR$(Q(25))+CHR$(Q(2
6))+CHR$(Q(27))+CHR$(Q(28))+CHR$(Q(29))+
CHR$(Q(30))+CHR$(Q(31))
1560 SPRITE$(1)=V$+W$+X$+Y$
1570 PUTSPRITE1,(208,19),15,1
1580 D=STICK(W):IFD=0THEN1580
1590 IFD=3THENM=1:SCREEN,3:GOTO1520
1600 IFD=7THENM=0:GOTO1500
1610 GOTO1610
1620 '---F1=SPRITE,F2=PRINT,F3=EDIT---
1630 ONKEYGOSUB1660,830,1280
1640 KEY(1)ON:KEY(2)ON:KEY(3)ON
1650 RETURN
1660 RETURN770
1670 ONKEYGOSUB,,1700
1680 KEY(3)ON
1690 RETURN
1700 RETURN630
1710 '---F1-F3=BACK TO MENU---
1720 ONKEYGOSUB1850,1850,1850
1730 KEY(1)ON:KEY(2)ON:KEY(3)ON
1740 SCREEN1,2:M=0:KEYOFF
1750 V$=CHR$(0)+CHR$(4)+CHR$(56)+CHR$(10
4)+CHR$(120)+CHR$(24)+CHR$(28)+CHR$(31)
1760 W$=CHR$(28)+CHR$(28)+CHR$(31)+CHR$(
24)+CHR$(24)+CHR$(24)+CHR$(24)+CHR$(24)
1770 X$=CHR$(0)+CHR$(0)+CHR$(0)+CHR$(0)+
CHR$(0)+CHR$(0)+CHR$(0)+CHR$(248)
1780 Y$=CHR$(118)+CHR$(113)+CHR$(240)+CH
R$(48)+CHR$(48)+CHR$(48)+CHR$(48)+CHR$(4
8)
1790 SPRITE$(1)=V$+W$+X$+Y$
1800 PUTSPRITE1,(100,100),,1
1810 D=STICK(W):IFD=0THEN1810
1820 IFD=3THENM=1:SCREEN2,3:GOTO1750
1830 IFD=7THENM=0:GOTO1740
1840 GOTO1840
1850 RETURN50
```

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